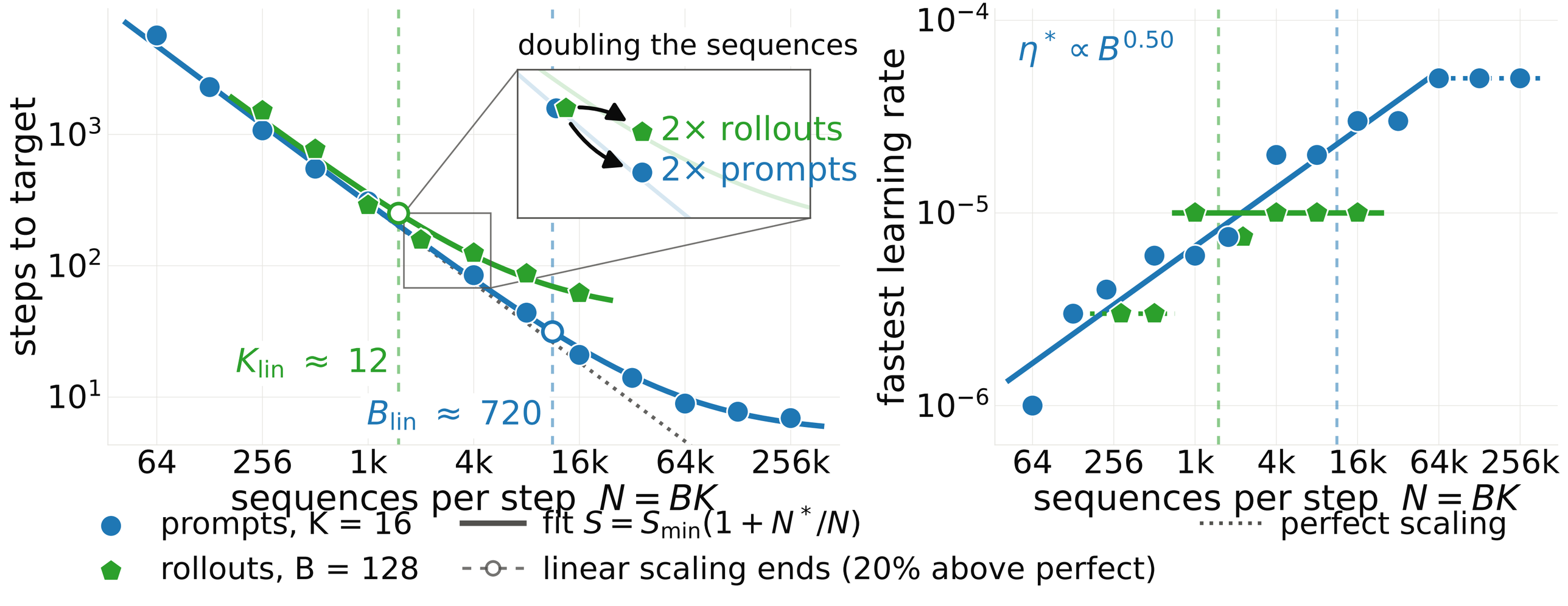
steps to target
10^3
10^2
10^1
doubling the sequences
2× rollouts
2× prompts
$K_{\text{lin}} \approx 12$
$B_{\text{lin}} \approx 720$
64
256
1k
4k
16k
64k
256k
sequences per step $N = BK$
fastest learning rate
10^{-4}
10^{-5}
10^{-6}
$\eta^* \propto B^{0.50}$
64
256
1k
4k
16k
64k
256k
sequences per step $N = BK$
prompts, K = 16
rollouts, B = 128
fit $S = S_{\min}(1 + N^*/N)$
linear scaling ends (20% above perfect)
perfect scaling